

Vijay Baskar

Lead Software Engineer @ Retailetics | AI/ML Developer | Mobile Systems & Scaled Engineering
+91 8088413589 | yoursvijii@gmail.com | linkedin.com/in/vijaybaskar

SUMMARY

Lead Software Engineer with 13+ years building production Android, backend, and on-device AI/ML systems. Hands-on with Kotlin, Java, Jetpack Compose, MVVM, REST APIs, Spring Boot, Firebase, TensorFlow, PyTorch, TensorFlow Lite, RAG, and MLOps. Reduced ANR crashes by 70%, improved app performance by 30%, designed Spring Boot microservices for 500K+ users (40% faster responses), and shipped AI-powered self-checkout and indoor navigation that saved over \$2M. Led design reviews, mentored junior engineers, and built CI/CD (Jenkins, Fastlane) with automated testing (JUnit, Espresso). Seeking AI, mobile, and backend engineering roles.

SKILLS

Programming Languages: Java, Kotlin, Dart, Python, MATLAB

Android / Mobile: Kotlin, Java, Android SDK, Jetpack Compose, Jetpack components, MVVM, Clean Architecture

Cross-Platform: Flutter, Dart, Kotlin Multiplatform (KMP), iOS, state management (BLoC, Provider, MVP)

Backend & APIs: REST / RESTful APIs, GraphQL, WebSockets, Spring Boot, microservices, Firebase, AWS, Postgres, SQLite

AI / ML: TensorFlow, TensorFlow Lite, PyTorch, Keras, OpenCV, on-device ML, edge ML, model quantization, computer vision

Generative AI & MLOps: RAG, agentic systems, tool-calling, LLM prompting, Transformers, training-to-serving pipelines, model serving, Jenkins, Docker, Git

Libraries & Data: NumPy, Pandas, Scikit-learn, NLTK, DNN, CNN, RNN, LSTMs, GANs, NLP (text pre-processing, sentiment analysis, language modeling)

DevOps & Testing: Jenkins, Fastlane, Bitrise, Git, GitHub, Bitbucket, GitLab, Docker, CI/CD, JUnit, Espresso, Flutter Test

Architecture: SOLID, Clean Architecture, MVVM, MVP, high-performance Android, indexing, caching, pagination

IoT & Hardware: Arduino, Raspberry Pi, Orange Pi, Bluetooth Beacons, IMU Sensors, sockets

Retail / In-store Systems: ezyCart, self-checkout kiosk, AI recipe builder, NFC / Tap to Pay, product finder, voice assist, stock audit, ezyList, Flutter SDK lite

EXPERIENCE

Retailetics SDN BHD — Lead Software Engineer | February 2021 – Present

Kuala Lumpur, Malaysia

- Spearheaded the AI-powered self-checkout smart trolley (ezyCart) on Android: system architecture, stakeholder design, and a real-time socket pipeline between IoT devices and the ezyCart Android app (Kotlin, WebSockets) for fast data sync without redundant permission requests.
- Shipped the self-checkout kiosk on the same stack (scan, bag, pay) and integrated payment gateways including Tap to Pay and NFC so checkout works on cart and kiosk.
- Built an on-cart AI recipe system: cuisine suggestions from basket items, cooking video, missing-product add-to-cart, plus in-aisle upsell and cross-sell ads.
- Delivered AI-powered indoor navigation and product finder (search a SKU, navigate the aisle), replacing thousands of Bluetooth beacons and saving over \$2M.
- Shipped stock audit on the smart cart, Voice Assist AI for product and promo queries, and monitoring/troubleshooting for every active cart and self-checkout station.
- Designed and developed a cross-platform Flutter support-staff app, an Android TV cart monitor, and a Flutter SDK lite of ezyCart for scan-and-pay in vendor phone and tablet apps.
- Shipped ezyList (shared lists, nearby store price/availability, navigate, one-scan purchase, digital receipt, member promos) and Lisa, an AI shopping assistant for live queries, recommendations, and promotions.
- Developed and deployed machine learning models; led automated build pipeline and deployment across merchants and outlets; led split testing initiatives that drove additional revenue.
- Optimized high-performance Android architecture (70% fewer ANRs), Spring Boot microservices for 500K+ users (40% faster responses), and database access (60% faster queries via indexing, caching, and pagination). Integrated on-device TensorFlow Lite models (quantization, ~30% faster inference) for computer vision and recommendation-style mobile AI.

Colan infotech PVT LTD — Senior Software Engineer | November 2012 – February 2021

Chennai, India

- Delivered 20+ production mobile apps (Android/iOS): native Kotlin/Java (Android SDK, MVVM, Jetpack Compose) and Flutter (state management, widgets, animations, integrations).
- Implemented CI/CD with Jenkins, GitLab, and Bitrise; architected scalable mobile backends using AWS, Firebase, and RESTful APIs.

Advanced AI/ML Engineering Project — Proprietary R&D · Forex market indicator

- Designed and trained models that read FX market structure and render an inspectable indicator overlay for research and analysis. Applied time-series and deep-learning methods (including on-device and quantized inference patterns used in production mobile work) in a sequential build: indicator, then a calibrated signal layer, then an execution prototype with explicit risk limits.

EDUCATION

Anna University / GGR College of Engineering — Bachelor of Engineering in Computer Science | Vellore, India | June 2007 – November 2011

Target Software Solution — Mobile Application Development using Android and iOS | Chennai, India | January 2012 – October 2012

Additional coursework: SOLID Principles — Coursera; AI/ML Architecture — IBM; Master prompting — Claude and Coursera